

1-8-2026

MCA-0303 Reinforcement Learning class Test-2 192425090

1. Reinforcement Learning (RL) is a type of machine learning where an agent learns by interacting with an environment.

Objective: To maximize the total cumulative reward by choosing the best actions.

2. MDP is a mathematical model used to solve decision-making problems in RL.

a. State (S)

b. action (A)

c. Reward (R)

d. Transition probability (P)

e. Discount factor (γ).

3. The Bellman equation is:

$$V(S) = R(S) + \gamma \sum P(S' | S, a) V(S')$$

$V(S)$ = State value

$R(S)$ = Reward

γ = Discount factor.

4.

Exploration:

- * Tries new actions.
- * gain new knowledge.

exploitation:

- * chooses the best known action.
- * maximizes reward using known knowledge.

5. The multi-Armed Bandit problem is choosing the best action from several options with unknown rewards. It balances exploration and exploitation to maximize reward.

6. Dynamic programming is a method used to solve complex problems by breaking them into smaller sub problems.

- policy evaluation
- policy improvement
- value iteration
- policy iteration

7. TD Learning is an RL method that updates value estimates using the current reward and estimated next state value.

TD(0):

$$V(S) = V(S) + \alpha [R + \gamma V(S') - V(S)]$$

α = Learning rate

R = Reward

γ = discount factor.

8.

SARSA

Φ -Learning

* on-policy algorithm

* off-policy

* uses the next chosen action

* uses the best possible action

* safer learning

* faster learning

9. DDPG is a deep learning-based RL algorithm that uses a neural network to estimate

* Handles large state spaces.

* Learns directly from images/features.

* Better performance than Φ -tables.

10. Δ QPV is an Improved version of QPV that reduces overestimation of p-values.

preferred because:-

- more accurate p-value estimation
- more stable learning.
- better performance than QPV.